

Andrew Peterson

QUANTUM RESEARCHER · SECURITY ENGINEER · SOFTWARE ARCHITECT

California, United States of America

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Summary

Software engineer with 4+ years building quantum computing platforms, cloud infrastructure, and full-stack applications. Co-founded UCSD's first quantum computing org (100+ members), engineered backends for 500+ participant global competitions, and built quantum-encrypted video systems at Qualcomm Institute. Seeking roles at the intersection of emerging technology and production-grade engineering.

Experience

Qualcomm Institute

La Jolla, California

QUANTUM SOFTWARE ENGINEERING LEAD (INTERN)

Jul. 2022 - Dec. 2024

- Architected a quantum-encrypted video platform via photonic circuitry, achieving secure key distribution at 8 qubits with <4.2% QBER over 1.2-meter channels.
- Optimized Grover's algorithm in Qiskit for binary constraint satisfaction, achieving 4x speedup over classical brute-force on 16-variable problems with 38% circuit depth reduction.
- Built and benchmarked QML classification models via quantum neural networks, achieving 94.7% accuracy on Iris, outperforming classical baselines by 2.3 percentage points.
- Mentored a cross-functional team of 4 interns under Dr. Nikola Alic, overseeing software deliverables and commissioning a new undergraduate quantum computing lab.

Rochester Institute of Technology Esports

Remote (Rochester, New York)

FULL-STACK DEVOPS ENGINEER & ADVISOR

Sep. 2020 - Jul. 2022

- Engineered and deployed MERN-stack web resources via Docker, serving 2,400 monthly active users across 12 varsity esports teams.
- Orchestrated cloud infrastructure on DigitalOcean Droplets, implementing Nginx reverse proxies and Docker Compose to ensure 99.9% uptime for high-traffic services.
- Built CI/CD pipelines and monitoring suites, reducing deployment time from 6 hours (manual) to 15 minutes and saving ~320 hours/year in release effort.

Extracurricular

Quantum Computing at UC San Diego

La Jolla, CA

CO-FOUNDER & PRESIDENT (2023 - 2024) & TECHNICAL CURRICULUM LEAD

Feb. 2021 - Dec. 2024

- Co-founded and scaled the first undergraduate quantum computing organization at UC San Diego, establishing a community for 100+ aspiring researchers and engineers.
- Developed a technical curriculum spanning 6 quarterly workshops, educating 200+ students with average attendance growing from 12 to 45.

San Diego Capture the Flag (SDCTF)

Remote

LEAD INFRASTRUCTURE ENGINEER & CHALLENGE DESIGNER

Dec. 2020 - May 2022

- Engineered the backend for San Diego's largest annual cybersecurity competition, supporting 500+ global participants in its inaugural year.
- Provisioned cloud-native infrastructure using Terraform (IaC), managing a Postgres database and containerized challenge environments.
- Designed and deployed a Discord.JS-integrated submission bot to automate real-time flag validation and leaderboard updates.

Skills

Languages

Python (Expert), C/C++, Java, TypeScript/JavaScript, SQL, Bash, LaTeX, Matlab, Rust

Systems & DevOps

Docker, Kubernetes, Terraform, Linux Kernel, Distributed Systems, CI/CD, Nginx, Git, AWS, DigitalOcean

Software Engineering

Data Structures & Algorithms (DSA), System Design, Design Patterns, Agile/Scrum, Unit Testing

Data Science & AI

PyTorch, Scikit-Learn, Pandas, Computer Vision, NLP, Ollama (LLMs), Statistical Modeling, Jupyter

Infrastructure

PostgreSQL, MongoDB, Redis, RESTful/GraphQL API Design, Microservices, Linux Server Administration

Certifications

2025 **Azure Data Fundamentals**, Microsoft

2022 **IT Fundamentals+**, CompTIA

Education

University of California, San Diego

La Jolla, California

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

Sep. 2020 - Dec. 2024

- Relevant Coursework: Advanced Data Structures, Design & Analysis of Algorithms, Operating Systems, Machine Learning, Computer Security, Networked Systems, Software Engineering, Discrete Mathematics